

Vulnerability Assessment Report – Database Server 1. Purpose of the Report

The objective of this vulnerability assessment is to identify and analyze the risks associated with the publicly exposed database server, which stores critical customer information and supports e-commerce operations. Protecting this data is essential to maintain customer trust, transaction integrity, and business continuity. The analysis aims to provide a detailed view of the vulnerabilities, allowing the company to implement appropriate security controls and mitigate the potential impacts of cyberattacks.

- | | | | |
|-------------------|--------------------------------------|--|---|
| 2. Threat Sources | Threat Source Description | External hackers | Malicious agents who may exploit public access to the server to steal data or disrupt services. |
| | Business competitors | May attempt to compromise strategic data or gain a competitive advantage using intrusion techniques. | |
| | Disgruntled or negligent employees | May cause accidental or deliberate data exposure through unsafe practices. | |
| 3. Threat Events | Threat Event | Threat Source | Probability (1-3) Severity (1-3) Risk (1-9) |
| | Exfiltration of customer data | External hackers | 3 3 9 |
| | Denial-of-service (DoS) attack | Business competitors | 2 2 4 |
| | Unauthorized access to internal data | Disgruntled/negligent employees | 2 3 6 |

4. Approach

The vulnerability assessment was conducted qualitatively, considering the probability and severity of attacks based on prior cybersecurity knowledge. The three selected threat sources represent the most critical risks to the company, including data loss, service interruption, and exposure to internal malicious actions. The analysis focused on the exposed database server, its attack surface, and the potential consequences for business operations and customers.

5. Remediation Strategy

To mitigate the identified risks, the following controls are recommended:

Least Privilege: Limit database access to authorized personnel only.

Multi-Factor Authentication (MFA): Require MFA for all remote server access.

Defense in Depth: Implement firewalls, network segmentation, and continuous monitoring.

AAA Framework: Centralized authentication, authorization, and auditing to track suspicious activity.

Public Key Infrastructure (PKI): Protect sensitive data in transit and at rest, preventing exfiltration.

These measures align with the risks of exfiltration, service interruption, and unauthorized access, improving the overall security posture of the server.